TECHNICAL INFORMATION LETTER

7F AND 9F AFT END COMPRESSOR RUBS

APPLICATION
All 7F and 9F Heavy Duty Gas Turbines

PURPOSE
To inform users to monitor equipment operation and condition, and to identify the onset, occurrence, and effects of aft end compressor rubs (when the tip of the rotor blade rubs the casing). Rubs are evident when blade tip material is adversely affected. Information in this TIL is supplemental to information in the Inspection and Maintenance Manual provided with the gas turbine.

REASON FOR REVISION
To update the technical content of the TIL, including frequency of tip loss and recommendations when discovered.

Compliance Category

O - Optional	Identifies changes that may be beneficial to some, but not necessarily all, operators. Accomplishment is at customer's discretion.
M - Maintenance	Identifies maintenance guidelines or best practices for reliable equipment operation.
C - Compliance Required	Identifies the need for action to correct a condition that, if left uncorrected, may result in reduced equipment reliability, efficiency, or safety. Compliance may be required within a specific operating time.
A - Alert / Safety	Failure to comply with the TIL could result in equipment damage, facility damage, or personal injury. Compliance is mandated within a specific operating time.

Timing Code

1	Prior to Unit Startup / Prior to Continued Operation (forced outage condition)
2	At First Opportunity (next shutdown)
3	Prior to Operation of Affected System
4	At First Exposure of Component
5	At Scheduled Component Part Repair or Replacement
6	Next Scheduled Outage
7	Optional

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BACKGROUND DISCUSSION

The compressor section of F-Class gas turbines has 18 (0-17) stages of rotating components (rotor blades) and stationary components (stator vanes).

Some F-class machines have experienced aft end compressor rubs. These rubs range from light scoring on the casing to material loss at the blade tip. This phenomenon was concluded to be very low risk relative to unit operation and reliability. No performance impact and no forced outages have been experienced as a result. Rotor blades in rows 13, 14, and 15 are primarily affected.

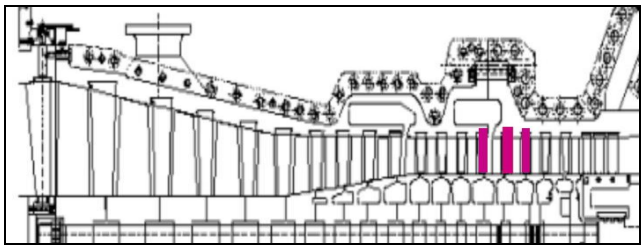


Figure 1: F-Class Compressor Cross-Section

Based on the conclusion of Root Cause Analysis (RCA) and fleet experience, aft end rubs are primarily induced because of frequent *restarts* between two (2) to eight (8) hours after shutdown. During this transient period, the thermal compression of the casing relative to the rotor can decrease clearances and can lead to rubs upon restart. The heat generated from a rub can lower the local material fatigue strength of the blade resulting in material tip loss in the event of another rub. Typical material loss dimensions are less than 1/16 inch^2 in area (see Figure 2).

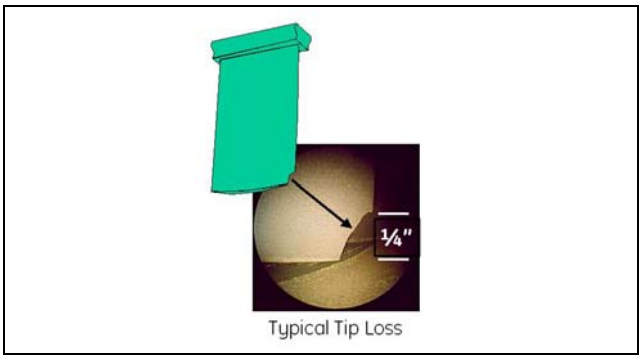


Figure 2: Typical Blade Tip Loss

Restarting a unit 2 to 8 hours after a shutdown or trip is within the operating envelope, but this may cause rubs in some cases. Aft end rubs and/or tip loss pose very low risk to unit operation and reliability. No performance impact and no forced outages have been experienced due to this issue.

RECOMMENDATIONS

Follow operating practices per GER 3620 to reduce the potential for rubs on aft end compressor blades. In addition to the GER 3620 guidelines, the following recommendations would also help minimize aft end rubs and tip loss occurrence.

- Keep enclosure doors closed during startup, shutdown and normal operation.
- Be aware that restarting the gas turbine between two (2) and eight (8) hours following a shutdown or trip will cause the clearances in the aft end to decrease. If the unit operates this way now or has in the past, there is a potential for this clearance closure to cause aft end tip rubs. As a result, the area should be included in the yearly borescope inspections.
- Eliminate any potential pipe leaks that may cause quenching of the gas turbine compressor discharge casing.
- Ensure vent fans and exhaust fans to the gas turbine enclosure are functioning properly such that no local forced convection is taking place at or near the compressor discharge casing.
- Integrate the inspection for aft end compressor rubs with your yearly scheduled borescope inspection per GER 3620. Observe rotor blade stages 13 through 17 and look for mushroomed material (see Figure 3), bluing at the blade tip, and / or markings in the casing area.

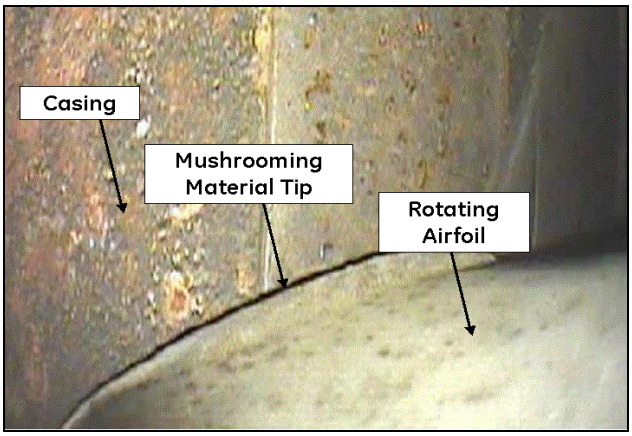


Figure 3: Fractured Tip Surface

If rubs or tip loss do occur, blending to increase clearances can be performed at Major Inspections (MI) or when the Compressor Discharge Casing (CDC) is removed. During repairs, clearances of rotor blades can be increased radially by tip grinding and polishing, and tip corners repaired by blade blending. Typically, 20 mils radially of material can be removed to both increase clearance and remove heat affected and rolled over material.

Following this operation, a penetrant inspection must be performed to ensure no tip cracks are present. If the CDC is removed prior to the major inspection, clearances can also be increased to eliminate the potential for tip rubs should the customer desire. Please contact your local GE service representative for assistance.

PLANNING INFORMATION

Compliance

- Compliance Category: **M**
- Timing Code: **6**

Manpower Skills

- Trained GE Borescope technician to perform borescope inspection, approximately eight (8) hours.
- Trained GE Blade technician for rotor blade clearance increases, approximately 16 hours once the casing has been removed.

Parts
N/A

Special Tooling
N/A

Reference Documents
N/A

Previous Modifications
N/A

Scope of Work
N/A

Contact your local GE Energy Services representative for assistance or for additional information.

NOTE: *If you would like to receive future TILs by email, contact your local GE Energy Services representative for assistance.*

TIL COMPLIANCE RECORD

Compliance with this TIL must be entered in local records. GE requests that the customer notify GE upon compliance of this TIL.

Complete the following *TIL Compliance Record* and FAX it to:

TIL Compliance
FAX: (678) 844-3451
Toll free FAX: 1-888-896-TILS (1-888-896-8457)

TIL COMPLIANCE RECORD	For Internal Records Only # _____
Site Name:	Customer Name:
Customer Contact Information	**GE Contact Information**
Contact Name:	Contact Name:
Address:	Address:
Email:	Email:
Phone:	Phone:
FAX:	FAX:
Turbine Serial Number(s):	

INSTALLED EQUIPMENT	TIL Completed Date: _____
	100% TIL Completed: _____
Description:	

Unit Numbers:	Part Description:	Part Number	MLI Number

Comments:

NOTE: *If there are any redlined drawings that pertain to this TIL implementation, please FAX the drawings along with this TIL Compliance Record.*

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Serial Number: _____ **Date:** _____

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1 - 5 days **6 - 10 days** **+ 10 days**

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Rate the following based on a scale of 1 to 5, where **1** is Excellent and **5** is Poor.

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3. Please rate the overall effectiveness of this TIL.

1 **2** **3** **4** **5**

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